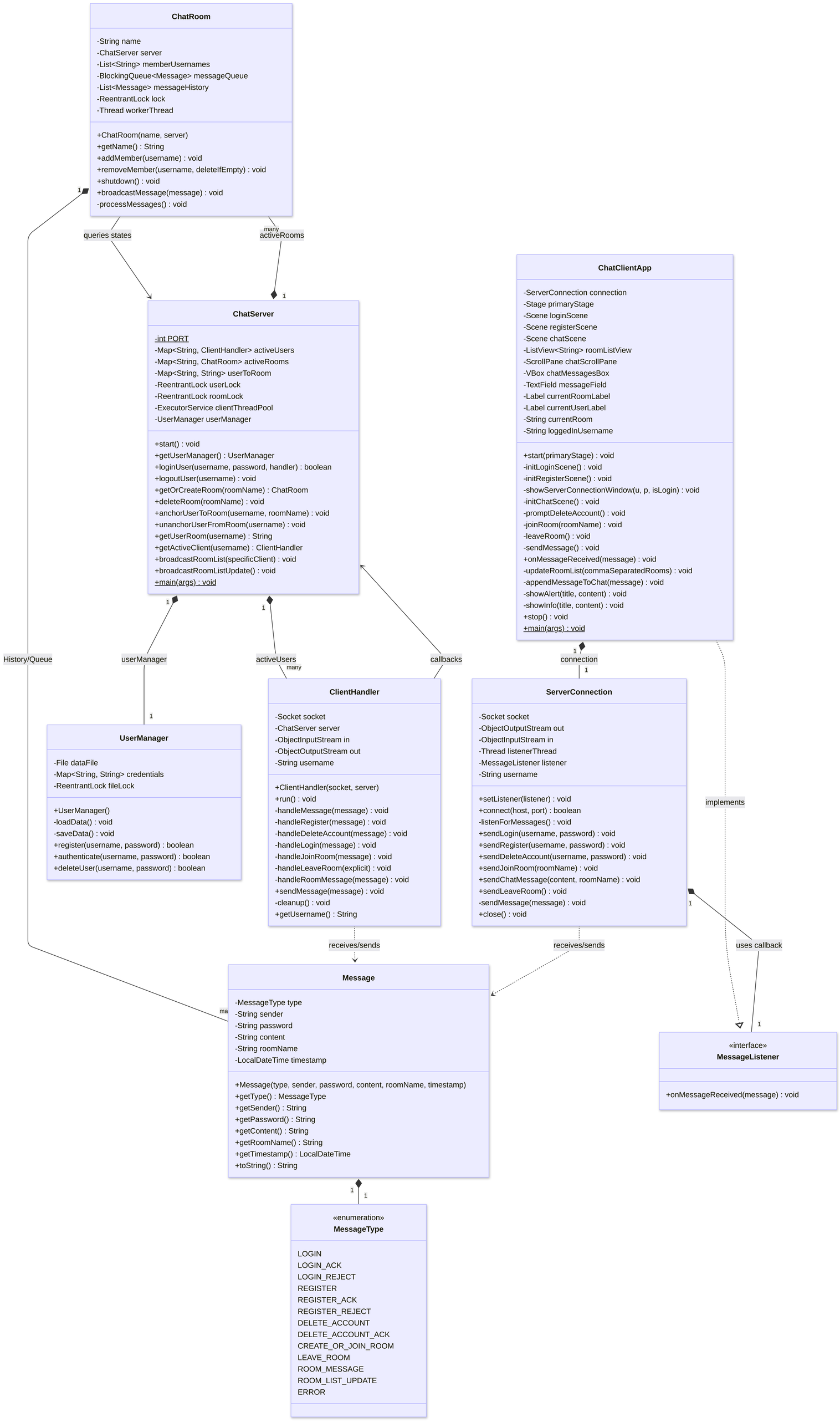
ChatRoom
-String name
-ChatServer server
-List<String> memberUsernames
-BlockingQueue<Message> messageQueue
-List<Message> messageHistory
-ReentrantLock lock
-Thread workerThread
+ChatRoom(name, server)
+getName() : String
+addMember(username) : void
+removeMember(username, deleteIfEmpty) : void
+shutdown() : void
+broadcastMessage(message) : void
-processMessages() : void
queries states
many
activeRooms
ChatServer
-int PORT
-Map<String, ClientHandler> activeUsers
-Map<String, ChatRoom> activeRooms
-Map<String, String> userToRoom
-ReentrantLock userLock
-ReentrantLock roomLock
-ExecutorService clientThreadPool
-UserManager userManager
+start() : void
+getUserManager() : UserManager
+loginUser(username, password, handler) : boolean
+logoutUser(username) : void
+getOrCreateRoom(roomName) : ChatRoom
+deleteRoom(roomName) : void
+anchorUserToRoom(username, roomName) : void
+unanchorUserFromRoom(username) : void
+getUserRoom(username) : String
+getActiveClient(username) : ClientHandler
+broadcastRoomList(specificClient) : void
+broadcastRoomListUpdate() : void
+main(args) : void
ChatClientApp
-ServerConnection connection
-Stage primaryStage
-Scene loginScene
-Scene registerScene
-Scene chatScene
-ListView<String> roomListView
-ScrollPane chatScrollPane
-VBox chatMessagesBox
-TextField messageField
-Label currentRoomLabel
-Label currentUserLabel
-String currentRoom
-String loggedInUsername
+start(primaryStage) : void
-initLoginScene() : void
-initRegisterScene() : void
-showServerConnectionWindow(u, p, isLogin) : void
-initChatScene() : void
-promptDeleteAccount() : void
-joinRoom(roomName) : void
-leaveRoom() : void
-sendMessage() : void
+onMessageReceived(message) : void
-updateRoomList(commaSeparatedRooms) : void
-appendMessageToChat(message) : void
-showAlert(title, content) : void
-showInfo(title, content) : void
+stop() : void
+main(args) : void
History/Queue
userManager
activeUsers
callbacks
connection
UserManager
-File dataFile
-Map<String, String> credentials
-ReentrantLock fileLock
+UserManager()
-loadData() : void
-saveData() : void
+register(username, password) : boolean
+authenticate(username, password) : boolean
+deleteUser(username, password) : boolean
ClientHandler
-Socket socket
-ChatServer server
-ObjectInputStream in
-ObjectOutputStream out
-String username
+ClientHandler(socket, server)
+run() : void
-handleMessage(message) : void
-handleRegister(message) : void
-handleDeleteAccount(message) : void
-handleLogin(message) : void
-handleJoinRoom(message) : void
-handleLeaveRoom(explicit) : void
-handleRoomMessage(message) : void
+sendMessage(message) : void
-cleanup() : void
+getUsername() : String
ServerConnection
-Socket socket
-ObjectOutputStream out
-ObjectInputStream in
-Thread listenerThread
-MessageListener listener
-String username
+setListener(listener) : void
+connect(host, port) : boolean
-listenForMessages() : void
+sendLogin(username, password) : void
+sendRegister(username, password) : void
+sendDeleteAccount(username, password) : void
+sendJoinRoom(roomName) : void
+sendChatMessage(content, roomName) : void
+sendLeaveRoom() : void
-sendMessage(message) : void
+close() : void
implements
receives/sends
receives/sends
uses callback
Message
-MessageType type
-String sender
-String password
-String content
-String roomName
-LocalDateTime timestamp
+Message(type, sender, password, content, roomName, timestamp)
+getType() : MessageType
+getSender() : String
+getPassword() : String
+getContent() : String
+getRoomName() : String
+getTimestamp() : LocalDateTime
+toString() : String
«interface»
MessageListener
+onMessageReceived(message) : void
«enumeration»
MessageType
LOGIN
LOGIN_ACK
LOGIN_REJECT
REGISTER
REGISTER_ACK
REGISTER_REJECT
DELETE_ACCOUNT
DELETE_ACCOUNT_ACK
CREATE_OR_JOIN_ROOM
LEAVE_ROOM
ROOM_MESSAGE
ROOM_LIST_UPDATE
ERROR